

Introduction to Multimedia

Multimedia means that computer information can be represented through audio, video, and animation in addition to traditional media (i.e., text, graphics/drawings, and images).

Multimedia is the field concerned with the computer controlled integration of text, graphics, and drawings, still and moving images (Video), animation, audio, and any other media where every type of information can be represented, stored, transmitted and processed digitally.

Multimedia is a media that uses multiple form of information content and information processing.

It is a platform of presenting. Multimedia can refer to all sorts' files including text files. The different types of multimedia are:

- Text and sound
- Text, sound, and still or animated graphic images
- Text, sound, and video images
- Video and sound
- Multiple display areas, images, or presentations presented concurrently
- In live situations, the use of a speaker or actors and "props" together with sound, images, and motion video

The basic elements of multimedia

- Text
- Graphics
- Animation
- Video
- Audio

TEXT

Characters that are used to create words, sentences, and paragraphs.

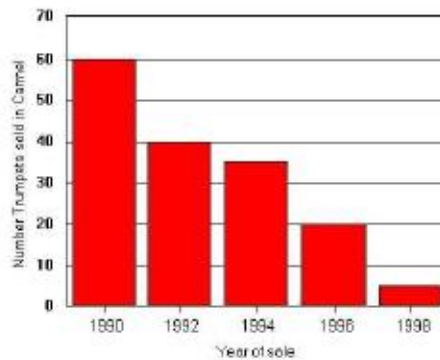
Titles

Multimedia is a rich medium that accommodates numerous instructional strategies. Multimedia addresses many of the challenges of instruction in both the academic and corporate environments. It is accessible over distance and time and provides a vehicle for consistent delivery. Multimedia can provide the best medium with which to communicate a concept.

- Monitor
- Keyboard
- Mouse
- Speaker

Graphics

A digital representation of non-text information, such as a drawing, chart, or photograph.



Animation

Flipping through a series of still images. It is a series of graphics that create an illusion of motion.



Video

Photographic images that are played back at speeds of 15 to 30 frames a second and they provide the appearance of full motion.



Audio

Music, speech, or any other sound.



Categories of Multimedia

Two types of Multimedia presentation

- Linear Presentation
- Non-linear Interactive

Linear active content progresses often without any navigational control for the viewer such as a cinema presentation.

Non-linear uses interactivity to control progress as with a video game or self-paced computer based training. Hypermedia is an example of non-linear content.

Application areas of Multimedia

Multimedia finds its application in various areas including, but not limited to:

- Advertisements
- Art
- Education

- Entertainment
- Engineering
- Medicine
- Mathematics
- Business
- Scientific research

In education, multimedia can be used as a source of information. Students can search encyclopedias such as Encarta, which provide facts on a variety of different topics using multimedia presentations.

Teachers can use multimedia presentations to make lessons more interesting by using animations to highlight or demonstrate key points.

A multimedia presentation can also make it easier for pupils to read text rather than typing to read a teacher's writing on the board.

Programs which show pictures and text whilst children are reading a story can help them learn to read; these too are a form of multimedia presentation.

Multimedia is used for advertising and selling products on the Internet.

Some businesses use multimedia for training where CDROMs or on-line tutorials allow staff to learn at their own speed, and at a suitable time to the staff and the company.

Another benefit is that the company do not have to pay the additional expenses of an employee attending a course away from the workplace.

People use the Internet for a wide range of reasons, including shopping and finding out about their hobbies.

The Internet has many multimedia elements embedded in web pages and web browsers support a variety of multimedia formats.

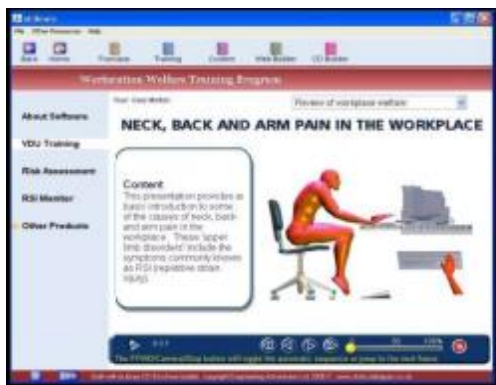
Many computer games use sound tracks, 3D graphics and video clips.



Computer-Based Training



Teaching Aid



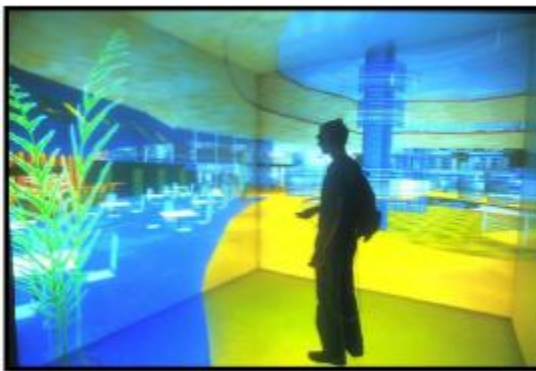
References



Entertainment



Simulation



Virtual Reality



Virtual Surgery



Information Kiosk

Advantages of using multimedia

It is very user-friendly. It doesn't take much energy out of the user, in the sense that you can sit and watch the presentation, you can read the text and hear the audio.

It is multi sensorial. It uses a lot of the user's senses while making use of multimedia, for example hearing, seeing and talking.

It is integrated and interactive. All the different mediums are integrated through the digitization process. Interactivity is heightened by the possibility of easy feedback.

It is flexible. Being digital, this media can easily be changed to fit different situations and audiences.

Loading multimedia components

```
import javax.swing.*;
import java.awt.*;
import java.net.*;
import javax.media.*;

public class Nepal extends javax.swing.JFrame {

    public Nepal(URL url) {
        initComponents();
        setVisible(true);
        try {
            Player p=Manager.createRealizedPlayer(new MediaLocator(url));
            Component visual=p.getVisualComponent();
```

```

        Component control=p.getControlPanelComponent();
        if(visual!=null){
            add(visual,BorderLayout.CENTER);
        }
        add(control,BorderLayout.SOUTH);
        p.start();
    } catch (Exception e) {
        System.out.println("Error "+e);
    }
}

private void initComponents() {

    setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

    javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(0, 400, Short.MAX_VALUE)
            )
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addGap(0, 300, Short.MAX_VALUE)
            )
    );

    pack();
}

public static void main(String args[]) {
    try {
        JFileChooser jf=new JFileChooser();
        jf.showOpenDialog(null);
        URL url=null;
        url=jf.getSelectedFile().toURI().toURL();
        Nepal obj=new Nepal(url);
    } catch (Exception e) {
        System.out.println("Error "+e);
    }
}
}

```